BONUS 5 · BAMBU STUDIO 2.8.2

Calibration Guide

What the printer measures by itself, the two wizards worth running, the first-layer check.

What runs automatically

The self-test and calibration the screen guides you through on first boot are the printer measuring itself. Loud vibrations during this stage are normal: part of the routine shakes the printer on purpose.

Routine	What it does	What you gain
Auto bed leveling	Measures the height of the plate surface at many points	An even first layer over small dips and rises
Nozzle height	Measures the exact gap between nozzle tip and plate	A first layer pressed on, not scraped or floating
Vibration compensation	Measures the printer's vibration resonance	Part of the automatic calibration Bambu runs on every new printer
Motor noise cancellation	Calibrates the motors to cut noise during long, fast moves	Less humming; slightly fewer fine vertical marks on walls

- Let the printer measure itself rather than leveling by hand. The A1 levels and sets its nozzle height automatically, with no paper test.
- The A1, A1 mini and P2S use an eddy current sensor to measure nozzle pressure and compensate the flow; this flow dynamics calibration can also run at the start of a print.
- Let every routine your screen lists finish. Some models add their own, such as the H2D's motion accuracy calibration or the X1 series' Micro-Lidar scans.
- To repeat a routine later, for example after belt work: the printer's menu, or your printer in Bambu Handy > **Settings** > **Calibration**.

> **WARNING**
> The bed and toolhead move fast and without notice during calibration. Keep hands, cords and loose sleeves out of the machine, and do not wear gloves or anything that could catch. Do not touch the nozzle or the heated bed while the printer is working.

In Send print job, for every print

Which calibration options show depends on the printer. Each has a switch with **Auto**, **On** and **Off**, or only **On** and **Off**.

- ☐ **Auto Bed Leveling**: leave it on **Auto** or **On**, not **Off**.
- ☐ **Flow Dynamics Calibration**: **On** for any filament that has no saved result yet, and whenever a case in the table below applies.

When each calibration is worth running

	Flow dynamics (pressure)	Flow rate (quantity)
Bambu filament	Recalibrate in the cases below	Usually unnecessary, Bambu says
Worth running when	A new filament; a new or worn nozzle; after drying damp filament; a changed temperature or **Max volumetric speed**	Other brands that show too much or too little plastic, such as overflowing top surfaces
Modes	Automatic or manual	Automatic on printers with the Micro-Lidar (X1 series); manual coarse, then fine
Limits	Unreliable on damp or transparent filament; runs vary by about 10%	Values above 1.00 are rarely needed
Result	A K value, stored per filament	The **Flow ratio** in the filament preset (0.98 for Bambu PLA Basic)

The **Calibration** tab appears for Bambu Lab printers only and holds two wizards, **Flow Dynamics** and **Flow Rate**. Each prints a small test, so Studio must reach the printer.

Flow Dynamics wizard

Before you start: the filament dried and loaded, with its type set; a clean plate whose type you know; the printer idle. Allow time for one short test print.

- [ ] Click the **Calibration** tab.
- [ ] Click **Flow Dynamics** in the list of wizards.
- [ ] Click **Auto-Calibration** or **Manual Calibration**.
- [ ] Choose the plate type.
- [ ] Choose the slot that holds the filament.
- [ ] Click **Calibrate**.
- [ ] Wait for the test to finish printing.
- [ ] Manual mode: pick the line that looks most even along its whole length.
- [ ] Enter its value where the wizard asks, then click **Next**.
- [ ] Save the result under a name that says which filament it belongs to, then click **Finish**.

It worked when: the result appears in the list of saved results under **Manage Result**, which lets you pick a K value per slot.

If not: lines hard to tell apart mean the filament may still be damp. Dry it and run the test again. Results within about 10% of each other are normal variation.

Flow Rate wizard

For: a filament from another brand. Skip it for Bambu filament. **Before you start:** as for Flow Dynamics. Allow time for two short test prints.

- [ ] Click the **Calibration** tab.
- [ ] Click **Flow Rate** in the list of wizards.
- [ ] Start the coarse pass: blocks from 80% to 120% in 5% steps.
- [ ] Wait for the blocks to finish printing.
- [ ] Pick the block with the smoothest top surface.
- [ ] Click the second **Calibrate**, on the page where you entered the coarse result, to start the fine pass: blocks from 91% to 100% of the coarse result in 1% steps.
- [ ] Wait for the blocks to finish printing.
- [ ] Pick the smoothest block again.
- [ ] Save the result.

It worked when: the new **Flow ratio** shows in the filament preset.

If not: blocks that all look alike point to damp filament. A filament that still struggles at speed may need a lower **Max volumetric speed** in its filament preset (Chapter 8).

First-layer check

On screen, before you send

- [ ] Click **Slice plate**.
- [ ] Drag the layer slider, at the right of **Preview**, down to layer 1.
- [ ] Press Ctrl+1 for the top view.
- [ ] Every part shows a solid first layer over its whole footprint, with the brim or ears you asked for.

Rerun auto bed leveling (Bambu Handy)

Printer idle, bound to Handy, clean plate on the bed.

1. Open Bambu Handy and tap the **Device** tab.
2. Tap your printer, then **Settings** at the top right.
3. Tap **Calibration** and select the auto bed leveling item.
4. Start the calibration and wait until the printer reports that it is finished.

On the printer, while the first layer goes down

What you see	Likely cause	First thing to do
Flat lines that touch, forming one even sheet	A good first layer	Nothing
Round strands with gaps, lines dragged behind the nozzle	Nozzle too high; for example, the wrong plate tile on textured PEI	Check the plate tile
Lines that hold in some areas and lift in others	Oil on the plate, often where fingers touched	The two-minute wash
Thin, rough lines with ridges between them	Nozzle too low, or the bed leveling is off	Check the nozzle tip (Chapter 13), then rerun auto bed leveling
Corners that rise after a few layers	Warping	Plate tile and wash first, then a brim or ears (Chapter 10)

Watch from outside the print area. Leave the Z offset to the printer: on a clean plate with the right tile, rerun the calibration instead of adjusting it by hand.